Name:_____

1 Series

When we express a number by an infinite decimal, such as $\pi = 3.14159265\ldots$, what we are really saying is it can be expressed as an infinite sum of the form

$$\pi = 3 + \frac{1}{10} + \frac{4}{10^2} + \frac{1}{10^3} + \frac{5}{10^4} + \frac{9}{10^5} + \frac{2}{10^6} + \frac{6}{10^7} + \frac{5}{10^8} + \cdots.$$

We have not yet defined the sum of infinitely many numbers. In the expression for π above, the dots mean that as we add more terms, then the sum gets closer and closer to the actual value of π. That is, we define the *infinite sum* above as the limit of the sequence of *partial sums*, if this limit exists. More generally, we can consider adding the terms of any infinite sequence $\{a_i\}_{i=1}^{\infty}$.

Definition 1.1 (Sum of a sequence)**.** Given a sequence $\{a_i\}_{i=1}^{\infty}$, define the *nth partial sum* of the sequence by $s_n = \sum_{i=1}^{n} a_i$. We will say that the sequence $\{a_i\}$ is *summable* if the sequence $\{s_n\}$ converges. In this case, $\lim_{n\to\infty} s_n$ is denoted by $\sum_{i=1}^{\infty} a_i$ and is called the *sum of the sequence* $\{a_i\}$. That is,

$$\sum_{i=1}^{\infty} a_i := \lim_{n\to\infty} \sum_{i=1}^{n} a_i,$$

provided this limit exists.

1. Find the nth partial sum of the sequence $\{i\}_{i=1}^{\infty}$ and use it to determine whether the sequence is summable.

2. Prove that if $\{a_i\}$ and $\{b_i\}$ are summable, then

$$\sum_{i=1}^{\infty}(a_i + b_i) = \sum_{i=1}^{\infty} a_i + \sum_{i=1}^{\infty} b_i,$$
$$\sum_{i=1}^{\infty} ca_i = c\sum_{i=1}^{\infty} a_i.$$

Remark 1.2. Unfortunately, no one uses the terminology in Definition 1.1. Instead, the expression $\sum_{i=1}^{\infty} a_i = a_1 + a_2 + a_3 + \cdots$ is called an *infinite series* (or just *series*). If $\{a_i\}_{i=1}^{n}$ is summable, then one says the series $\sum_{i=1}^{\infty} a_i$ *converges*. In this case, one calls $s = \lim_{n\to\infty} s_n$ the *sum of the series*, and one writes $\sum_{i=1}^{\infty} a_i = s$. When $\{a_i\}$ is not summable, one says the series $\sum_{i=1}^{\infty} a_i$ *diverges*.

This terminology is rather strange, because at best the expression $\sum_{i=1}^{\infty} a_i$ denotes a number (so it can't "converge"), and it doesn't denote anything at all unless $\{a_i\}$ is summable. Nevertheless, this terminology is standard and we will use it going forward.

2 Geometric Series

An important example of an infinite series is the *geometric series* $\sum_{n=1}^{\infty} ar^{n-1}$, $a \neq 0$.

2. Find the nth partial sum of the geometric series. For $r \neq 1$, the standard trick is to let $s_n = \sum_{i=1}^{n} ar^{i-1} = a + ar + ar^2 + \cdots + ar^{n-1}$, then form the difference $s_n - rs_n$ and solve for s.

3. Using your expression in Problem 2, calculate $\lim_{n\to\infty} s_n$. For which values of r does this limit exist? Find the sum of the series when it exists.

♢ Check your answers to Problems 2 and 3 with your instructor before moving on.

4. Determine whether the geometric series $5 - \frac{10}{3} + \frac{20}{9} - \frac{40}{27} + \cdots$ converges. If so, find the sum.

5. Determine whether the series $\sum_{i=1}^{\infty} 2^{2n} 3^{1-n}$ converges.

6. Write the number $2.3\overline{17} = 2.3171717\ldots$ as a ratio of integers.

3 Telescoping Series

7. Consider the series $\sum_{i=1}^{\infty} \frac{1}{i(i+1)}$.

   (a) Use partial fractions to rewrite $\frac{1}{i(i+1)}$ in the form $\frac{1}{i(i+1)} = \frac{A}{i} + \frac{B}{i+1}$.

   (b) Use your answer to part (a) to find the nth partial sum s_n of the series.

   (c) Use your answer to part (b) to find $\lim_{n\to\infty} s_n$.

Name:_____

8. Find the sum of each of the following series.

(a) $\sum_{i=2}^{\infty} \frac{1}{i^2-1}$

(b) $\sum_{i=1}^{\infty} \frac{8}{(4i-3)(4i+1)}$

Name:_____

(c) $\sum_{k=0}^{\infty} (\arctan(k+1) - \arctan(k))$

(d) $\sum_{j=1}^{\infty} \ln\left(\frac{j+1}{j}\right)$